



AI Agent Development Internship

at ScaleOn

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Program & Semester: B.Tech CSE, 7th Semester

Organization: ScaleOn

OVERVIEW

Agenda

- About ScaleOn & the Internship Program
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- Project 3 — ScaleOn Study Portal
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- My Contributions & Learnings
- Conclusion

INTRODUCTION

About the Internship

- Role: AI Agent Developer Intern at ScaleOn (Intern ID: SO260037)
- Duration: 01 July 2026 – 30 September 2026
- ScaleOn builds AI-powered agentic software products under the tagline "Scale Beyond Limits"
- Worked across three live projects spanning multi-agent AI, social media automation, and study-portal platforms
- Focus areas: agentic workflow design, backend engineering, LLM integration, and frontend development

PROJECT 01

ScaleProposal AI

AI-powered multi-agent business proposal generation platform

- Decomposes proposal generation into 4 specialized agents: Planner → Pricing → Risk → Proposal
- Each agent uses strict Pydantic schema contracts to pass structured data downstream
- Real-time progress via Server-Sent Events (SSE), isolated per workflow
- Persistent storage with SQLAlchemy + Alembic, client-side PDF export
- Built with Next.js frontend + FastAPI backend + Google Gemini AI

Agent Pipeline

1. PlannerAgent — scope & roadmap
2. PricingAgent — cost estimation
3. RiskAgent — risk assessment
4. ProposalAgent — final document

PROJECT 02

ScaleSocial AI

Enterprise multi-channel social media management suite

- Unified batch composer — post once to Twitter/X, LinkedIn, Facebook & Instagram
- Gemini 2.5 AI Writer Engine for platform-optimized copy with iterative refinement
- Analytics dashboard: impressions, engagement rate, reactions, live tweet metrics
- Visual publishing calendar with scheduled content and saved drafts
- Zero-trust security: JWT auth via Supabase, Redis-backed rate limiting

Tech Stack

- Next.js 15 + TypeScript
- FastAPI (Python)
- Celery + Redis workers
- Supabase (Auth + Postgres)
- Gemini 2.5 AI

PROJECT 03

ScaleOn Study Portal

- A study/learning management portal built with a documented backend and frontend architecture
- REST API layer (documented in `API_DOCUMENTATION.md`) connecting frontend to backend services
- Structured database design for managing courses, study material, and student data
- Includes architecture, database, and changelog documentation for maintainability

Technologies & Tools Used

Frontend

- Next.js 15 (App Router)
- React + TypeScript
- Zustand state mgmt
- Tailwind CSS

Backend

- Python + FastAPI
- Pydantic schemas
- Celery workers
- Alembic migrations

AI & Data

- Google Gemini AI
- SQLAlchemy ORM
- SQLite / PostgreSQL
- Supabase, Redis

Quality

- Pytest / Pytest-asyncio
- Mock AI providers
- JWT authentication
- Git & GitHub

MY ROLE

My Contributions

- Contributed to agentic AI pipeline design across the Planner–Pricing–Risk–Proposal workflow
- Built and tested FastAPI backend endpoints with Pydantic schema validation
- Integrated Google Gemini AI for content generation and structured reasoning
- Worked on Next.js frontend components for real-time workflow interaction
- Wrote automated Pytest test cases using mock AI providers and in-memory fixtures
- Maintained project documentation (README, architecture notes) for handover clarity

REFLECTION

Key Learnings

- Practical experience designing multi-agent AI systems with structured schema contracts
- Hands-on exposure to production-grade FastAPI + Next.js full-stack architecture
- Understanding of secure authentication, rate limiting, and background job processing
- Improved skills in API integration with LLMs (Google Gemini) for real-world use cases
- Learned to work in an agile, documentation-driven engineering environment

WRAP-UP

Conclusion

This internship at ScaleOn provided valuable, hands-on experience in building agentic AI systems and full-stack applications, from architecture design to implementation and testing across three distinct products. The exposure to real production codebases, modern AI integration, and collaborative engineering practices has strengthened my practical skills as an AI Agent Developer and prepared me for future software engineering roles.

Thank You